Structure-Adaptive Convolutional Neural Network for Hyperspectral Image Classification

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***Abstract*— Hyperspectral image (HSI) classification based on deep learning is a hot research topic. The convolutional model employs a single rectangular window to interpret the sample neighborhood features, whereas effective characterization of the complex spatial structure of HSI is still an unsolved problem. In this article, we propose a structure-adaptive convolutional neural network (SACNN) for HSI classification, which efficiently exploits the intrinsic spatial geometry information. Four novel strategies are designed to construct the proposed SACNN network. First, superpixel homogeneous region (SHR) sample generation is introduced to achieve neighborhood features within the intercepted rectangular window of the superpixel. Second, online batch-wise standardization uses zero padding to unify the size of inputs in the same batch, thereby realizing parallel processing of irregular inputs. Third, structure-adaptive convolution (SConv) and structure-adaptive average pooling (SAP) are correspondingly constructed to extract deep spectral, spatial, and geometric features from the effective mapping area of superpixels, and further aggregate the information within irregular boundaries. Finally, a sample-adaptive loss weight (SLW) scheme is designed to adjust the influence of different labels on the same input. Experimental results show that the overall classification accuracy of SACNN reaches 93.11%, 90.96%, and 85.04% for 15 randomly selected training samples per class on three HSI datasets, respectively, obtaining an improvement of 0.97%–2.97% with respect to the best-compared method.**

***Index Terms*— Convolutional neural network (CNN), hyperspectral image (HSI) classification, superpixel segmentation.**

I. Introduction

HYPERSPECTRAL image (HSI) contains rich spectral information reflecting the unique physical properties of ground objects. From the perspective of spectral dimension, each band can be regarded as an independent grayscale image. Alternatively, from the perspective of spatial dimension, each pixel can be viewed as a separate spectral response curve. In particular, HSI has the ability to realize attribute analysis and accurate characterization of target objects, which has been widely adopted in agricultural monitoring [1], [2] and environmental protection [3], [4]. HSI classification has always been one of the hottest research topics in HSI processing. However, excessively high spectral dimension brings information redundancy and at the same time faces the problem of limited labeled samples, which will easily lead to the performance degradation of the classifier and "Hughes phenomenon" [5], [6], [7].

HSI classification based on traditional machine learning extract features according to domain knowledge and send them to support vector machine (SVM) [8], random forest (RF) [9], K-nearest neighbor (KNN) [10], and other classifiers for identification. Clearly, discriminative feature extraction according to HSI characteristics is one of the keys to obtaining high classification performance. A series of linear methods such as linear discriminant analysis (LDA) [11], nonlinear methods such as kernel principal component analysis (KPCA) [12], and spatial-spectral kernel methods [13], are proposed for feature mapping. Furthermore, texture and morphological features are also widely used in the literature, such as extended morphological attribute profile (EMAP) [14]. Wavelet transform is employed for spatial-spectral information extraction [15], [16], [17]. In addition, a large number of spatial-spectral-based feature extraction methods [18], [19] have been proposed, simultaneously extracting spectral variation information and spatial distribution patterns.

Recently, deep learning has attracted much attention due to its excellent performance [20]. Unlike traditional machine learning with artificial feature extraction, deep-learning models extract features and recognize ground objects through an end-to-end network structure for HSI classification, such as autoencoder (AE) [21], [22], recurrent neural network (RNN) [23], convolutional neural network (CNN) [24], [25], attention network [26], [27], graph convolutional network (GCN) [28], and generative adversarial network (GAN) [29], [30], [31]. It is worth mentioning that the CNN model is most widely employed, due to its convolution calculation with local perception, weight sharing, and translation invariance. Zhao and Du [32] designed a 2DCNN to extract deep spatial features. Paoletti et al. [33] proposed a 3DCNN to process HSI in both spatial and spectral dimensions, attaining

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higher classification results than 2DCNN. However, the model has more parameters, resulting in low operating efficiency. Zhang et al. [34] leveraged the combination of different scale information from DenseNet to realize the coarse-to-fine feature extraction of HSI.

The deep-learning model often relies on a large number of labeled samples to achieve a stable convergence state and optimal classification performance [35]. In order to alleviate the conflict between limited labeled samples in HSI and training requirements, a variety of prior knowledge is integrated into the CNN model to improve feature discriminability and reduce sample dependence. Existing studies have shown that the combination of superpixel segmentation technology and CNN model can effectively enhance the spatial extraction ability and reduce the number of model parameters, thereby improving the performance of hyperspectral classification [36], [37]. Superpixel segmentation represents the shape and distribution of ground objects, which is utilized to optimize the preliminary classification results [38]. Superpixels have been integrated with CNN in the dual-window framework [39], 3-D architecture [40], ensemble learning approach [41], and multiscale adaptive model [42]. However, these methods fail to fully utilize the spatial partition information of superpixel homogeneous region (SHR), which is irregular and adheres to the contours of ground objects. They process the irregular superpixels in the form of fixed-sized rectangular windows during CNN training, which may miss internal information about large superpixels or introduce external interference for small superpixels. Common convolution and pooling operators based on regular patches are not suitable for irregular and uneven superpixel structures. Therefore, the irregular boundaries of superpixels should be considered comprehensively in CNN computation and further construct the internal spatial geometric structure of HSI.

To alleviate the above problems, we propose a structure-adaptive CNN (SACNN) for HSI classification, which is shown in Fig. 1. The SACNN framework mainly contains the modules of SHR sample acquisition, structure-adaptive convolution (SConv), structure-adaptive average pooling (SAP), and sample-adaptive loss weight (SLW). It helps to extract internal homogeneous information according to the irregular boundaries of superpixels, thus avoiding interference from external heterogeneous information. Specifically, first, simple linear iterative clustering (SLIC) superpixel segmentation algorithm [43] is adopted to over-segment the principal components of HSI. Second, the bounding rectangular windows (BRWs) of SHR are represented. Next, uniform size processing and regular zero padding are performed on the changing BRWs in batches online, so as to unify the window scales input in the same batch. Third, SConv and SAP calculations are designed to extract CNN intermediate layer features in the superpixel effective mapping area. Finally, the degree of influence of different labels on the same input is adjusted through SLW, and the flexibility and accuracy of object classification can be improved. In summary, the main contributions of this article are summarized as follows.

1) First of all, an SConv operator is proposed, which can not only automatically identify the effective mapping area of superpixels according to the distribution of eigenvalues but also analyze the intrinsic spatial-spectral structure information through geometric layer-by-layer feature extraction. Meanwhile, the depth features of the effective area can be retained while avoiding the interference of invalid area information. Moreover, there is no need to store each layer of the mapping area, and the amount of model parameters and training burden can be ensured.
2) Second, an SAP step is introduced to utilize the depth feature statistics of the entire superpixel effective mapping area. It can aggregate advanced spatial-spectral features within irregular boundaries, and enhance feature discrimination and saliency on the basis of SConv. In addition, 1×1 convolution is adopted to reduce the complexity and calculation amount of the model.
3) Third, in order to provide batch computing support for SConv and pooling, an SHR with irregular boundaries is employed as the input of the CNN model. The input size is normalized in batches, while the parallel processing of irregular superpixels is realized by unifying the input size of the same batch. Thus, the spatial division and information of superpixel segmentation can be fully utilized.
4) Finally, SLW calculation is designed, which can adjust the influence weight of different labels on the same input according to the spatial distribution of ground objects and the internal characteristics of superpixels, achieving the regularization effect to prevent overfitting. Furthermore, it jointly makes use of the prior knowledge of superpixel segmentation and spatial-spectral features learning and obtains optimal classification performance than other comparative methods in the experiments.

The rest of this article is organized as follows. Section II briefly introduces the related work on SLIC and CNN. The proposed SACNN framework is presented in detail in Section III. The HSI datasets used in this study and the experimental setup and results are provided in Section IV. The conclusion is summarized in Section VI.

II. Related Works

A. Simple Linear Iterative Clustering

Superpixel segmentation can reflect the spatial structure of the object and obtain a series of homogeneous regions, so it is widely favored in feature extraction of HSI. SLIC [43] performs local iterative clustering based on color similarity and spatial distance. It initializes the image as N_P superpixel clusters, and calculate the distance measure D in $2Q \times 2Q$ square region around each center, where $Q = \sqrt{\Sigma/N_P}$, Σ is the number of pixels in the image, then the distance between clustering center i and pixel in the block j can be expressed as

$$D_{i,j} = \sqrt{\left(\frac{D_{i,j}^{(S)}}{Q}\right)^2 + \left(\frac{D_{i,j}^{(C)}}{\varsigma_i}\right)^2} \tag{1}$$

where $D^{(S)}$ and $D^{(C)}$ are spatial and CIELAB color distance, respectively, and ς_i is the maximal value in the corresponding

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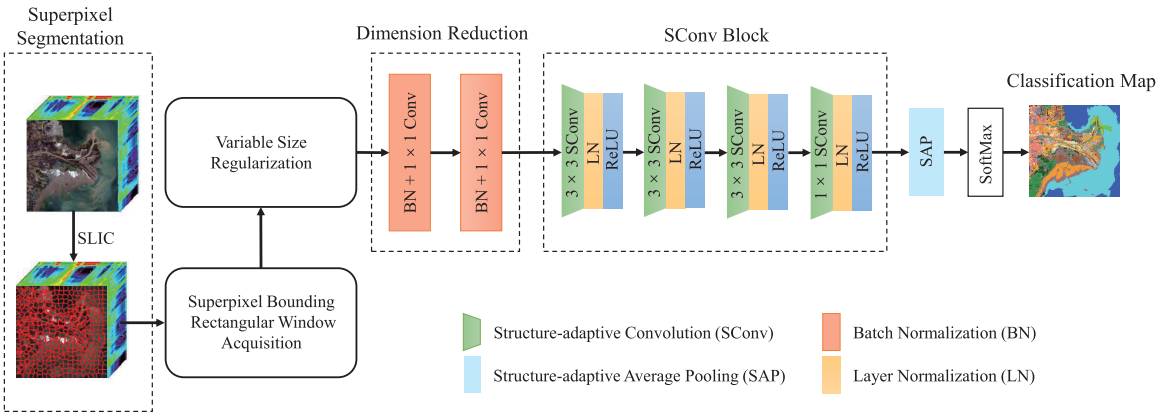


Fig. 1. Overview of the proposed SACNN framework for HSI classification.

group. (x, y) and (μ, ν, ω) denote the spatial position of the pixel and the CIELAB color value, respectively, then the spatial distance and color distance can be calculated by

$$D_{i,j}^{(S)} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} \tag{2}$$

$$D_{i,j}^{(C)} = \sqrt{(\mu_i - \mu_j)^2 + (\nu_i - \nu_j)^2 + (\omega_i - \omega_j)^2}. \tag{3}$$

The iteration continues until the clustering center no longer changes or the maximum number of iterations is reached. Finally, the superpixel segmentation result is obtained by merging the clusters that are too small or too similar.

B. CNN-Based Methods With Superpixel Segmentation

As a feed-forward neural network, CNN is widely adopted in various tasks of image and video analysis, natural language processing, and recommendation systems. In the task of HSI classification, HSI typically has high-dimensional features and limited labeled samples, but CNN-based classification methods often require high computational costs. To address this issue, researchers have explored the use of superpixel techniques to improve the classification performance of CNN models. Acción et al. [39] combined superpixels and CNN in a dual-window framework to decrease the training and prediction time for HSI classification, incorporating four different data augmentation techniques. In [40], a 3DCNN architecture was employed to simultaneously extract spatial-spectral information of HSI using superpixels to improve the classification performance, especially for noisy classification and boundary misclassification. More recently, Sikakollu and Dash [41] integrated multiple CNN classifiers and superpixel smoothing to remove the misclassified pixels and reduce the salt and pepper noise. The two dual-Channel CNN models explored both local binary patterns and HSI data simultaneously. Ren et al. [42] proposed a multiscale adaptive convolution model with superpixels, introducing adaptive kernels and adaptive convolution units to capture spatial-spectral features from various receptive fields of HSI. Xie et al. [44] embedded superpixel pooling operation in CNN with transfer learning to preserve the spatial structure information in the feature map.

However, the existing CNN-based methods with superpixels fail to fully utilize the spatial partition information of SHRs. They usually acquire the original hyperspectral cube with a fixed-size rectangular window centered on a given superpixel sample and utilize the obtained feature cube as model input. When a given sample is at the edge of the spatial area of land cover, the object type of feature cube cannot readily be determined by its center superpixel, because the cropped area likely belongs to the heterogeneous area and contains the information of different categories. Therefore, using the rectangular window input with a fixed size and a single shape for superpixels may introduce external interference or miss internal structure, due to ignoring the shape and distribution of ground objects. In addition, common convolution and pooling operators based on regular patches are not suitable for irregular and uneven superpixels. In irregular superpixels, the positions and distributions of edge grids cannot be simply mapped to square convolution kernels and pooling neighborhoods. When performing conventional convolution and pooling operations on irregular superpixels, signals beyond the edges of superpixels may be introduced, which can lead to information confusion and consequently affect classification accuracy. Based on the above analysis, we propose a SACNN for HSI classification.

III. Proposed SACNN Approach

Existing CNN-based methods with superpixels use fixed-size rectangular windows to calculate the HSI cubes with conventional convolution and pooling operators, which may introduce heterogeneity information and increase the difficulty of model training. As superpixel segmentation can divide the HSI into homogeneous regions and provide spatial structure information, we propose the SACNN method to effectively exploit the irregular superpixels and utilize the geometric information. There are four novel strategies in the SACNN framework: 1) SHR sample acquisition, which captures the superpixel neighborhood features and standardizes the input size of each batch, allowing parallel processing of irregular superpixels; 2) SConv, which extracts the deep spectral, spatial, and geometric features from superpixel-aligned samples; 3) SAP, which aggregates the extracted features within irregular boundaries of superpixels; and 4) SLW scheme, which optimizes the effect of different labels on the same sample.

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A. Sample Acquisition of SHR

1) SLIC Superpixel Over-Segmentation: Superpixels segmented by SLIC usually have good boundary fit and internal aggregation, which can effectively support the above hypothesis, that is, superpixel representation contains pixels belonging to the same category as the rectangular window. In order to ensure the homogeneous of each region, we apply SLIC on the original HSI data $\mathbf{H} \in \mathbb{R}^{X\times Y\times B}$ to obtain the over-segmented superpixel map. The number of superpixels N_{sp} is given as

$$N_{\text{sp}} = \frac{XY}{LW} \tag{4}$$

where X and Y are the length and width of the image spatial dimension, respectively. B is the number of spectral bands. L and W are the length and width of the general rectangular window input. Let the ith superpixel obtained by SLIC denote as $\mathcal{S}_i = \{\boldsymbol{s}_{i,1}, \ldots, \boldsymbol{s}_{i,j}, \ldots, \boldsymbol{s}_{i,N_p}\} \in \mathbb{R}^{N_p\times B}, i = 1, 2, \ldots, N_{\text{sp}}$, where vector $\boldsymbol{s}_{i,j} \in \mathbb{R}^{1\times B}$ is the feature vector corresponding to the jth pixel in the ith superpixel, and N_p is the number of pixels in the ith superpixel. Taking the upper left corner as the origin in spatial dimensions, the horizontal and vertical coordinates of each pixel are represented by the binary group (x, y), then the horizontal and vertical coordinates corresponding to the jth pixel in the ith superpixel are $(x_{s_{i,j}}, y_{s_{i,j}})$.

2) Superpixel BRW Extraction: After obtaining the SLIC superpixel over-segmentation, it is necessary to process different sizes of superpixel windows and irregular shapes to suit the end-to-end CNN model. One of the most direct ideas is to design a deformable convolution that can adapt to any input shape, but its design is complex, and the computational cost is huge. In order to solve the above problem, it is appropriate to regularize the BRW of each superpixel, satisfying the rectangular requirement of CNN input while maintaining spatial geometric structure. It is worth noting that the information belonging to the corresponding superpixel in the BRW is retained, and the part that does not belong to the superpixel has zero padding. In addition, in order to ensure the original spatial-spectral information of HSI is unchanged, only the BRW of the superpixel is considered. Fig. 2 shows a superpixel window and its BRW region. Let the BRW of the ith superpixel be $\boldsymbol{V}_i \in \mathbb{R}^{L_{\mathcal{S}_i}\times W_{\mathcal{S}_i}\times B}$, where $L_{\mathcal{S}_i} = \max_j x_{s_{i,j}} - \min_j x_{s_{i,j}}$ and $W_{\mathcal{S}_i} = \max_j y_{s_{i,j}} - \min_j y_{s_{i,j}}$, $j = 1, 2, \ldots, N_p$ denote its length and width, respectively. Then, the input cube $\boldsymbol{V}_i$ obtained by the ith superpixel can be expressed as

$$\boldsymbol{V}_i = \begin{bmatrix} \boldsymbol{v}_{i,1,1} & \cdots & \boldsymbol{v}_{i,1,W_{\mathcal{S}_i}} \\ \vdots & \ddots & \vdots \\ \boldsymbol{v}_{i,L_{\mathcal{S}_i},1} & \cdots & \boldsymbol{v}_{i,L_{\mathcal{S}_i},W_{\mathcal{S}_i}} \end{bmatrix} \tag{5}$$

$$\boldsymbol{v}_{i,l,w} = \begin{cases} \boldsymbol{s}_{i,j}, & \text{if } x_{\boldsymbol{v}_{i,l,w}} = x_{\boldsymbol{s}_{i,j}},\quad y_{\boldsymbol{v}_{i,l,w}} = y_{s_{i,j}},\ j \in [1, N_p] \\ 0, & \text{otherwise} \end{cases} \tag{6}$$

where vector $\boldsymbol{v}_{i,l,w}$ denotes the feature vector at the lth row and wth column in the ith superpixel.

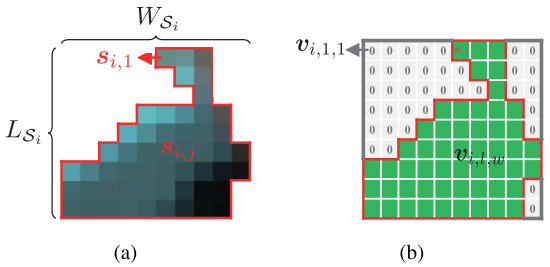


Fig. 2. Example of obtaining BRW $\boldsymbol{V}_i$ of the ith superpixel: (a) ith superpixel $\mathcal{S}_i$ and (b) BRW $\boldsymbol{V}_i$.

B. Structure-Adaptive Convolution

Through the above steps, the zero padding BRW based on SHR can be obtained. However, these different sizes of rectangular windows do not meet the requirement of batch training and iterative optimization in the CNN model. Therefore, it is necessary to design a structure-adaptive CNN model to extract spectral, spatial, and geometric depth features of irregular superpixels. In this section, we first consider the variable size regularization required for parallel operations and then introduce a SConv and SAP operation for the regular input. In addition, a SLW scheme is incorporated to adjust the influence of different labels on the same input.

Let $N_{\mathcal{T}}$ denote the total number of labeled samples, and $(x_{\mathcal{T}}^{(n)} y_{\mathcal{T}}^{(n)})$ denote the horizontal and vertical coordinates of the nth labeled sample. The input of the CNN model is expressed as the binary group $(\mathbf{F}_{\mathcal{T}}^{(n)}, \hat{\mathbf{F}}_{\mathcal{T}}^{(n)})$ including the feature map $\mathbf{F}_{\mathcal{T}}^{(n)} \in \mathbb{R}^{L_{\mathcal{T}}^{(n)}\times W_{\mathcal{T}}^{(n)}\times B}$ and its corresponding label $\hat{\mathbf{F}}_{\mathcal{T}}^{(n)} \in \mathbb{R}^1$, where $L_{\mathcal{T}}^{(n)}$ and $W_{\mathcal{T}}^{(n)}$ denotes the length and width of the feature map spatial dimension of the nth sample in the dataset $\mathcal{T}$, then all labeled samples can be expressed as a set $\mathcal{T} = \{(\mathbf{F}_{\mathcal{T}}^{(n)}, \hat{\mathbf{F}}_{\mathcal{T}}^{(n)})\}_{n=1}^{N_{\mathcal{T}}}$. For the SHR input, the sample feature map $\mathbf{F}_{\mathcal{T}}^{(n)}$ is the feature cube intercepted by the sample under the SHR. If $x_{\mathcal{T}}^{(n)} = x_{s_{i,j}}$ and $y_{\mathcal{T}}^{(n)} = y_{s_{i,j}}$, $i = 1, 2, \ldots, N_{\text{sp}}$, $j = 1, 2, \ldots, N_p$, then

$$\mathbf{F}_{\mathcal{T}}^{(n)} = \begin{bmatrix} \boldsymbol{f}_{\mathcal{T}\ 1,1}^{(n)} & \cdots & \boldsymbol{f}_{\mathcal{T}\ 1,W_{\mathcal{T}}^{(n)}}^{(n)} \\ \vdots & \ddots & \vdots \\ \boldsymbol{f}_{\mathcal{T}\ L_{\mathcal{T}}^{(n)},1}^{(n)} & \cdots & \boldsymbol{f}_{\mathcal{T}\ L_{\mathcal{T}}^{(n)},W_{\mathcal{T}}^{(n)}}^{(n)} \end{bmatrix} \tag{7}$$

$$\boldsymbol{f}_{\mathcal{T}\ l,w}^{(n)} = \boldsymbol{v}_{i,l,w} \tag{8}$$

$$L_{\mathcal{T}}^{(n)} = L_{\mathcal{S}_i} \tag{9}$$

$$W_{\mathcal{T}}^{(n)} = W_{\mathcal{S}_i} \tag{10}$$

where $\boldsymbol{f}_{\mathcal{T}\ l,w}^{(n)} \in \mathbb{R}^B$ denotes the feature vector in the lth row and the wth column of the feature map of the nth sample.

1) Variable Size Regularization: In the implementation of mini-batch gradient descent, in order to facilitate parallel processing and speed up training, samples in the same batch should have the same dimension, and superpixels with different sizes need to be regularized. Fig. 3 shows the schematic of the regularization operation for the samples in the same batch. $N_{\mathcal{T}_t}$ training samples are selected each time, where samples of the tth iteration are denoted as $\mathcal{T}_t = \{(\mathbf{F}_{\mathcal{T}_t}^{(n)}, \hat{\mathbf{F}}_{\mathcal{T}_t}^{(n)})\}_{n=1}^{N_{\mathcal{T}_t}} \in \mathcal{T}$, the maximum length $L_{\mathcal{T}_t} = \max_n L_{\mathcal{T}_t}^{(n)}$ and maximum width $W_{\mathcal{T}_t} = \max_n W_{\mathcal{T}_t}^{(n)}$, $n = 1, 2, \ldots, N_{\mathcal{T}_t}$ of

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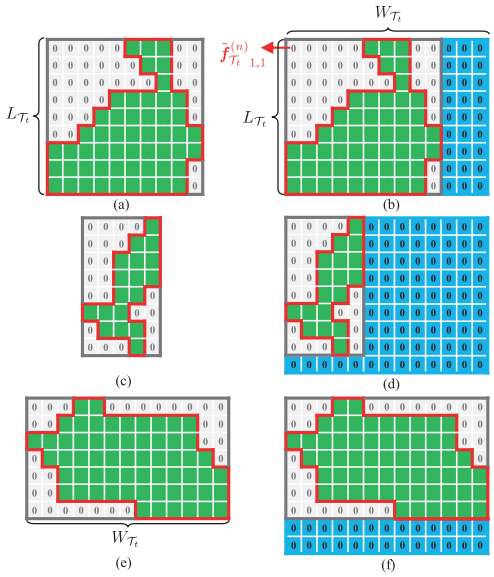


Fig. 3. Regularization of input in variable size: (a) feature map $\mathbf{F}^{(1)}_{\mathcal{T}_t}$ before regularization, (b) $\tilde{\mathbf{F}}^{(1)}_{\mathcal{T}_t}$ after regularization, (c) $\mathbf{F}^{(2)}_{\mathcal{T}_t}$ before regularization, (d) $\tilde{\mathbf{F}}^{(2)}_{\mathcal{T}_t}$ after regularization, (e) $\mathbf{F}^{(3)}_{\mathcal{T}_t}$ before regularization, and (f) $\tilde{\mathbf{F}}^{(3)}_{\mathcal{T}_t}$ after regularization.

the same batch of feature maps are adopted for the regular size. So regular feature map $\tilde{\mathbf{F}}^{(n)}_{\mathcal{T}_t} \in \mathbb{R}^{L_{\mathcal{T}_t} \times W_{\mathcal{T}_t} \times B}$ is calculated by

$$\tilde{\mathbf{F}}^{(n)}_{\mathcal{T}_t} = \begin{bmatrix} \tilde{\boldsymbol{f}}^{(n)}_{\mathcal{T}_t\ 1,1} & \cdots & \tilde{\boldsymbol{f}}^{(n)}_{\mathcal{T}_t\ 1,W_{\mathcal{T}_t}} \\ \vdots & \ddots & \vdots \\ \tilde{\boldsymbol{f}}^{(n)}_{\mathcal{T}_t\ L_{\mathcal{T}_t},1} & \cdots & \tilde{\boldsymbol{f}}^{(n)}_{\mathcal{T}_t\ L_{\mathcal{T}_t},W_{\mathcal{T}_t}} \end{bmatrix} \tag{11}$$

$$\tilde{\boldsymbol{f}}^{(n)}_{\mathcal{T}_t\ l,w} = \begin{cases} \boldsymbol{f}^{(n)}_{\mathcal{T}_t\ l,w}, & \text{if } l \le L^{(n)}_{\mathcal{T}_t} \text{ and } w \le W^{(n)}_{\mathcal{T}_t}, \\ & n \in [1, N_{\mathcal{T}_t}] \\ 0, & \text{otherwise} \end{cases} \tag{12}$$

where $\boldsymbol{f}^{(n)}_{\mathcal{T}_t\ l,w} \in \mathbb{R}^B$ and $\tilde{\boldsymbol{f}}^{(n)}_{\mathcal{T}_t\ l,w} \in \mathbb{R}^B$ denote feature vectors in the lth row and wth column of the nth sample in the tth batch $\mathcal{T}_t$ before and after regularization. Clearly, the regular superpixel feature map $\tilde{\mathbf{F}}^{(n)}_{\mathcal{T}_t}$ can be directly employed as the input of the CNN model. However, the above series of zero padding operations will inevitably bring interference information to the feature extraction. Therefore, calculations such as standard convolution and pooling need to be improved for superpixel-based feature map inputs.

2) Structure-Adaptive Convolution: The traditional convolution kernel and receptive field of the CNN model are set to be square. If the convolution operation is directly performed on the above-mentioned zero padding area, the geometric structure information of SHR will be lost, which could penalize spatial information mining and ground object feature extraction. In our SACNN approach, a SConv operation has been proposed to incorporate the prior knowledge of superpixel segmentation.

In the traditional convolution operation, the weight of the convolution kernel is shared, which means it cannot differentiate between pixels inside and outside the superpixel. When applied directly to the entire range using zero padding,

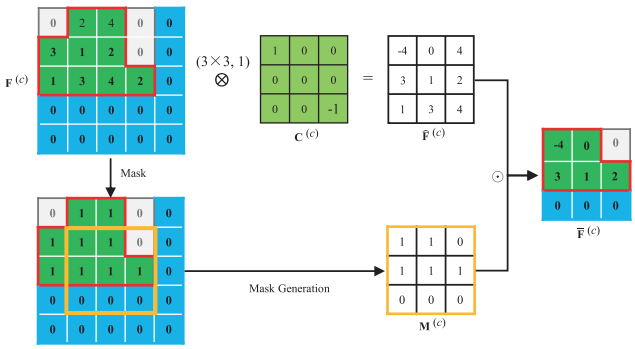


Fig. 4. Schematic of the proposed SConv, where ⊗ denotes 2-D convolution operation and ⊙ is element-wise multiplication.

this can result in the feature map containing invalid values outside the superpixel. To address these issues, we introduce a mask layer calculation to eliminate the invalid values in the feature map. The key difference between SConv and traditional convolution is that the former can adaptively adjust the weight of the convolution kernel based on the input content. This allows for better adaptation to the feature extraction requirements of different pixel positions. The ability of SConv to learn discriminative features relies on its boundary adaptability and content awareness. It enables the convolution kernel to be more flexible in adapting to the feature expression of different regions in the feature map. The specific calculation of the proposed SConv consists of two parts: 2-D convolution and mask generation, as shown in Fig. 4.

Let $\mathbf{F}^{(c)} \in \mathbb{R}^{L_{\mathbf{F}^{(c)}} \times W_{\mathbf{F}^{(c)}} \times K_{\mathbf{F}^{(c)}}}$ (when $c = 1$, $\mathbf{F}^{(1)} \in \{\tilde{\mathbf{F}}^{(n)}_{\mathcal{T}_t}\}_{n=1}^{N_{\mathcal{T}_t}}$) denote the input feature map of the cth ($c = 1, 2, \ldots, N_{\text{Sconv}}$) SConv layer and $\hat{\mathbf{F}}^{(c)} \in \mathbb{R}^{L_{\hat{\mathbf{F}}^{(c)}} \times W_{\hat{\mathbf{F}}^{(c)}} \times K_{\hat{\mathbf{F}}^{(c)}}}$ denote the output feature map of the traditional convolution kernel. Then, the 2-D convolution filter of this layer is denoted as $\mathbf{C}^{(c)} \in \mathbb{R}^{L_{\mathbf{C}^{(c)}} \times L_{\mathbf{C}^{(c)}} \times K_{\mathbf{C}^{(c)}}}$, where $L_{\mathbf{C}^{(c)}} \in \{3, 5, \ldots\}$ and $K_{\mathbf{C}^{(c)}} = K_{\mathbf{F}^{(c)}}$ denote the spatial size and the number of channels of the convolution filter, respectively. Meanwhile, $L_{\mathbf{F}^{(c)}}$, $W_{\mathbf{F}^{(c)}}$, $K_{\mathbf{F}^{(c)}}$ and $L_{\hat{\mathbf{F}}^{(c)}}$, $W_{\hat{\mathbf{F}}^{(c)}}$, $K_{\hat{\mathbf{F}}^{(c)}}$ are the length, width, and number of channels of the input and output feature maps, respectively. According to the convolution principle, given the above-mentioned convolution filter $\mathbf{C}^{(c)}$, the input feature map $\mathbf{F}^{(c)}$ is padded with zero, and $P^{(c)} \in \{0, 1, \ldots, L_{\mathbf{C}^{(c)}} - 2\}$ zeros are padded in each surrounding region. Then, the process of convolution with a step size $\text{Step}^{(c)}$ can be expressed as

$$\hat{\mathbf{F}}^{(c)} = \mathbf{F}^{(c)} \otimes \mathbf{C}^{(c)} \tag{13}$$

$$L_{\hat{\mathbf{F}}^{(c)}} = \frac{L_{\mathbf{F}^{(c)}} - L_{\mathbf{C}^{(c)}} + 2P^{(c)}}{\text{Step}^{(c)}} + 1 \tag{14}$$

$$W_{\hat{\mathbf{F}}^{(c)}} = \frac{W_{\mathbf{F}^{(c)}} - W_{\mathbf{C}^{(c)}} + 2P^{(c)}}{\text{Step}^{(c)}} + 1 \tag{15}$$

where ⊗ denotes a 2-D convolution operation.

Similarly, let $\mathbf{M}^{(c)} \in \mathbb{R}^{L_{\mathbf{M}^{(c)}} \times W_{\mathbf{M}^{(c)}}}$ be the mask and $\bar{\mathbf{F}}^{(c)} \in \mathbb{R}^{L_{\bar{\mathbf{F}}^{(c)}} \times W_{\bar{\mathbf{F}}^{(c)}} \times K_{\bar{\mathbf{F}}^{(c)}}}$ be the output feature map of the structure-adaptive convolutional layer, where $L_{\mathbf{M}^{(c)}} = L_{\hat{\mathbf{F}}^{(c)}}$ and $W_{\mathbf{M}^{(c)}} = W_{\hat{\mathbf{F}}^{(c)}}$ are the length and width of the mask, respectively. $L_{\bar{\mathbf{F}}^{(c)}} = L_{\hat{\mathbf{F}}^{(c)}}$, $W_{\bar{\mathbf{F}}^{(c)}} = W_{hat\mathbf{F}^{(c)}}$, and $K_{\bar{\mathbf{F}}^{(c)}} = K_{\hat{\mathbf{F}}^{(c)}}$ are the length, width, and number of channels of the output feature map of the structure-adaptive convolutional

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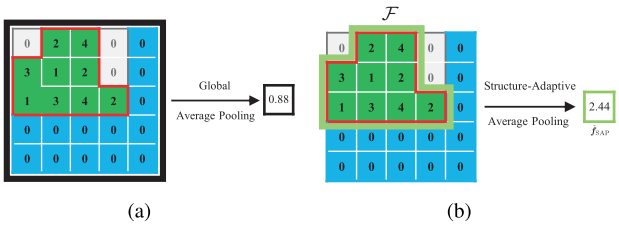


Fig. 5. Comparison of (a) global average pooling and (b) SAP.

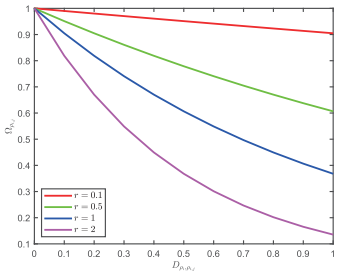


Fig. 6. Distance-weight ($D_{\rho_i,\rho_{i,j}} - \Omega_{\rho_{i,j}}$) curves with different r values.

layer, respectively. Fig. 4 shows the schematic of the SConv operation on the input feature map. Each position of the mask layer corresponds to the center of the sliding window convolution kernel in the input feature map. When performing a convolution operation, if the pixel in the input feature map corresponding to the center of the convolution kernel is nonzero, the corresponding position in the mask layer is 1. Conversely, the value is 0. Concerning the mask layer, it only contains two values 0 and 1, where 1 is related to the effective area of the original superpixel, and its corresponding convolution output is retained in the final feature map. Zero denotes an invalid region, which is removed in the final feature map. The generation of the mask is expressed as

$$\mathbf{M}^{(c)} = \begin{bmatrix} m_{1,1}^{(c)} & \cdots & m_{1,W_{\mathbf{M}^{(c)}}}^{(c)} \\ \vdots & \ddots & \vdots \\ m_{L_{\mathbf{M}^{(c)}},1}^{(c)} & \cdots & m_{L_{\mathbf{M}^{(c)}},W_{\mathbf{M}^{(c)}}}^{(c)} \end{bmatrix} \tag{16}$$

$$m_{l,w}^{(c)} = \begin{cases} 1, & \text{if } \boldsymbol{f}_{l,w}^{(c)} \neq 0, \quad l \leq L_{\mathbf{M}^{(c)}}, \ w \leq W_{\mathbf{M}^{(c)}} \\ 0, & \text{otherwise} \end{cases} \tag{17}$$

where $m_{l,w}^{(c)} \in \mathbb{R}^1$ denotes the lth row and wth column element and $\boldsymbol{f}_{l,w}^{(c)}$ denotes the corresponding vector of the input feature map of this layer. The mask provides the SConv calculation on the internal and external pixels of the superpixel, and the multiplication operation can preserve the convolution results of the effective area and eliminate the influence of the invalid area, achieving the superpixel-based SConv operation. The output feature map $\bar{\mathbf{F}}^{(c)} = \{\bar{\mathbf{F}}_k^{(c)}\}_{k=1}^{K_{\bar{\mathbf{F}}^{(c)}}}$ of the SConv layer is a modification of the traditional convolution output feature map $\hat{\mathbf{F}}^{(c)} = \{\hat{\mathbf{F}}_k^{(c)}\}_{k=1}^{K_{\hat{\mathbf{F}}^{(c)}}}$ by mask $\mathbf{M}^{(c)}$, that is, each channel $\hat{\mathbf{F}}_k^{(c)}$ is element-wise multiplied with the mask $\mathbf{M}^{(c)}$, which is

$$\bar{\mathbf{F}}_k^{(c)} = \hat{\mathbf{F}}_k^{(c)} \odot \mathbf{M}^{(c)}, \quad k = 1, 2, \ldots, K_{\hat{\mathbf{F}}^{(c)}} \tag{18}$$

where $\odot$ denotes the element-wise multiplication.

C. Structure-Adaptive Average Pooling

As the transformation of the convolutional layer becomes more flexible, the role of the pooling layer is weakened, and network design tends to use more convolutional layers to achieve comprehensive and in-depth feature learning. In addition, in the output stage of the network, global average pooling is employed to replace the fully connected layer, which can effectively reduce parameters and avoid model overfitting. For the superpixel regular window of the SACNN model, zero padding has a big impact on local area information and global average pooling. Here, a structural-adaptive average pooling (SAP) operation is designed, which is just applied to the SHRs.

Traditional global pooling performs average operation on the entire range including zero padding, which is affected by the invalid zero values outside superpixel, and fails to accurately reflect the valid information inside superpixel. Alternatively, SAP only averages pixel convolution results inside the superpixel by identifying the effective area of the superpixel and avoiding the impact of zero padding on the average pooling calculation. Fig. 5 shows a schematic of the SAP operation and traditional global average pooling on a single-channel input feature map.

Let the input and output feature maps of SAP be $\mathbf{F}_{\text{SAP}} \in \mathbb{R}^{L_{\mathbf{F}_{\text{SAP}}} \times W_{\mathbf{F}_{\text{SAP}}} \times K_{\mathbf{F}_{\text{SAP}}}}$ and $\hat{\mathbf{F}}_{\text{SAP}} \in \mathbb{R}^{K_{\hat{\mathbf{F}}_{\text{SAP}}}}$, where the input of the adaptive pooling layer is the output of the last adaptive convolutional layer $\mathbf{F}_{\text{SAP}} = \bar{\mathbf{F}}^{(N_{\text{SConv}})}$. $L_{\mathbf{F}_{\text{SAP}}}$, $W_{\mathbf{F}_{\text{SAP}}}$, and $K_{\mathbf{F}_{\text{SAP}}}$, respectively, denote the length, width, and number of channels of the input feature map of the adaptive pooling layer. The number of input channels and output dimensions must be equal to the number of sample label categories, that is, $K_{\mathbf{F}_{\text{SAP}}} = K_{\hat{\mathbf{F}}_{\text{SAP}}} = N_{\text{Class}}$. In addition, the pixel feature set corresponding to the original SHR in the input feature map is expressed as $\mathcal{F} \in \mathbb{R}^{L_{\mathcal{F}} \times K_{\mathcal{F}}}$, where $L_{\mathcal{F}}$ denotes the number of pixels that meets the condition, and $K_{\mathcal{F}} = N_{\text{Class}}$. The effective area of superpixels is chosen and mean operation is performed on internal pixels, so the superpixel structure can be aggregated and provide semantic features for subsequent recognition. In detail, SAP operation can be calculated as

$$\mathcal{F} = \boldsymbol{f}_{\text{SAP}l,w} | \boldsymbol{f}_{\text{SAP}l,w} \neq 0 \tag{19}$$

$$\hat{\boldsymbol{f}}_{\text{SAP}k} = \frac{1}{L_{\mathcal{F}}} \sum_{l=1}^{L_{\mathcal{F}}} \mathcal{F}_{l,k}, \quad k = 1, 2, \ldots, N_{\text{Class}} \tag{20}$$

where $\boldsymbol{f}_{\text{SAP}l,w}$ denotes the corresponding feature vector of the lth row and wth column element in the input feature map, and $\mathcal{F}_{l,k}$ denotes the feature value of the kth channel and the lth element in the original SHR. $\hat{\boldsymbol{f}}_{\text{SAP}k}$ denotes the kth eigenvalue of the SAP output feature map $\hat{\boldsymbol{f}}_{\text{SAP}}$.

D. Sample-Adaptive Loss Weight

Although SHR obtained by superpixel segmentation has good boundary fit and internal aggregation, there is still confusion at the boundary of the superpixel map. Using the above-mentioned superpixel regularization window, the same

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TABLE I
SACNN NETWORK STRUCTURE PARAMETERS

Operation	Convolution	Stride	Padding	Input	Output
SConv+LN+ReLU	$64@3 \times 3$	1	1	$L_{\mathcal{T}_t} \times W_{\mathcal{T}_t} \times B$	$L_{\mathcal{T}_t} \times W_{\mathcal{T}_t} \times 64$
SConv+LN+ReLU	$128@3 \times 3$	1	1	$L_{\mathcal{T}_t} \times W_{\mathcal{T}_t} \times 64$	$L_{\mathcal{T}_t} \times W_{\mathcal{T}_t} \times 128$
SConv+LN+ReLU	$256@3 \times 3$	1	1	$L_{\mathcal{T}_t} \times W_{\mathcal{T}_t} \times 128$	$L_{\mathcal{T}_t} \times W_{\mathcal{T}_t} \times 256$
SConv+LN+ReLU	$N_{\text{Class}}@1 \times 1$	1	0	$L_{\mathcal{T}_t} \times W_{\mathcal{T}_t} \times 256$	$L_{\mathcal{T}_t} \times W_{\mathcal{T}_t} \times N_{\text{Class}}$
SAP+SoftMax	-	-	-	$L_{\mathcal{T}_t} \times W_{\mathcal{T}_t} \times N_{\text{Class}}$	N_{Class}

feature map could correspond to multiple category labels, which could deteriorate network training and classification. In addition, the classical neural network usually assigns the same weight to all samples during training, while sample weighting or label smoothing is often employed to reduce the influence of sample distribution in the training of unbalanced datasets. Therefore, an SLW scheme is proposed to solve the problem of the same input corresponding to different labels and investigate the prior knowledge of superpixel segmentation as well.

First, let ρ_i denote the clustering center of the ith superpixel, its spatial position is (x_{ρ_i}, y_{ρ_i}), and the CIELAB color is $(\mu_{\rho_i}, \nu_{\rho_i}, \omega_{\rho_i})$. Suppose $\rho_{i,j}$ is the jth sample located in the ith superpixel, its spatial position is $(x_{\rho_{i,j}}, y_{\rho_{i,j}})$, and the CIELAB color is $(\mu_{\rho_{i,j}}, \nu_{\rho_{i,j}}, \omega_{\rho_{i,j}})$. According to (1)–(3), the distance between sample and its superpixel cluster center is $D_{\rho_i,\rho_{i,j}}$, and a heuristic manner is presented to make the sample loss weight adaptive to an appropriate value. From the principle of SLIC superpixel segmentation, the closer samples are to the superpixel clustering center, the larger the distribution of the corresponding category. That is to say, the category of samples near the center is more likely to be the one corresponding to the superpixel region. Hence, a larger weight is assigned to the sample loss function near the superpixel clustering center, whereas a smaller weight is assigned to the sample loss function far away from the center so that the training process can purposely strengthen the label with high probability and weaken the label with low probability. The calculation process of SLW $\Omega_{\rho_{i,j}}$ is as follows:

$$\Omega_{\rho_{i,j}} = \mathrm{e}^{-r \times D_{\rho_i,\rho_{i,j}}} \tag{21}$$

where parameter r is employed to control the degree of loss weight decay. If r is too large, the network training may not converge, while if r is too small, it will be unable to address the issue of different labels for the same input. Distance-weight $(D_{\rho_i,\rho_{i,j}} - \Omega_{\rho_{i,j}})$ curve for different values of r is shown in Fig. 6. It is observed that as the distance $D_{\rho_i,\rho_{i,j}}$ between sample and cluster center increases, the decay rate of sample loss weight $\Omega_{\rho_{i,j}}$ is diverse under different r values. Specifically, when r is large, the sample loss weight $\Omega_{\rho_{i,j}}$ decays sharply as $D_{\rho_i,\rho_{i,j}}$ increases, and vice versa. In this article, a moderate decay rate of sample loss weight as $D_{\rho_i,\rho_{i,j}}$ increases is more appropriate for the loss calculation and the label decision, so r is set to 0.5.

E. SACNN Architecture

The architecture of the proposed SACNN model is shown in Fig. 1, and its detailed network parameters are described in Table I. It consists of three 3×3 SConv layers, a 1×1 SConv layer, and an SAP layer. The number of convolution filters in SConv layers increases layer by layer so as to gradually extract the deep features of spectral, spatial, and geometric structures of irregular superpixels. The 1×1 SConv layer is employed for the dimension reduction of feature maps, reducing the number of channels to match the number of categories (N_{Class}) and adapting to the final output dimension. The SAP layer aggregates the deep feature information from the effective area of superpixels and connects the extracted feature maps to the category label distribution. Moreover, instead of using batch normalization (BN) [45], layer normalization (LN) [46] is employed in the network to adjust the distribution within each layer. This is because BN may not be suitable for small batch inputs with different effective areas, whereas LN normalizes each sample individually without being affected by the size of the effective area between samples.

The cross-entropy loss function is commonly used in deep neural networks for multiclassification problems to measure model errors and aid in parameter updates. Therefore, a cross-entropy loss function with SLW is designed in SACNN to quantify the discrepancy between model predictions and real labels. Different from the cross-entropy loss function, the loss function of SACNN is defined as

$$\text{Loss}_{sa} = -\sum_{\mathbf{F}_{\mathcal{T}_t}^{(n)} \in \mathcal{T}_t} \Omega_{\mathbf{F}_{\mathcal{T}_t}^{(n)}} \boldsymbol{p}_r\left(\mathbf{F}_{\mathcal{T}_t}^{(n)}\right) \log \boldsymbol{p}_\theta\left(\mathbf{F}_{\mathcal{T}_t}^{(n)}\right) \tag{22}$$

where $\mathbf{F}_{\mathcal{T}_t}^{(n)} \in \mathcal{T}_t$ denotes the superpixel regularized feature map of a single training sample, $\Omega_{\mathbf{F}_{\mathcal{T}_t}^{(n)}}$ denotes the loss weight of the sample, $\boldsymbol{p}_r(\cdot)$ and $\boldsymbol{p}_\theta(\cdot)$ are the real label distribution of sample and the predicted probability distribution by SoftMax layer, respectively, encoded by one-hot $\boldsymbol{p}_r(\mathbf{F}_{\mathcal{T}_t}^{(n)}) = \hat{\mathbf{F}}_{\mathcal{T}_t}^{(n)}$ in vector form.

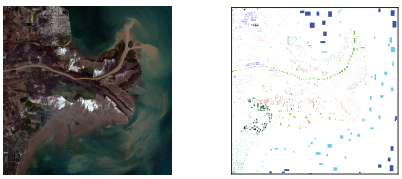
To summarize, the SACNN algorithm flow mainly consists of the following steps, as shown in Algorithm 1. First, the HSI data $\mathbf{H}$ is preprocessed using the SLIC superpixel segmentation, and zero padding is performed on the superpixels to obtain their respective BRWs $\boldsymbol{V}_i$. The regularized feature maps $\widetilde{F}_{\mathcal{T}_t}^{(n)}$ are obtained by regularizing the BRWs. Then, in the training phase, the regularized feature maps undergo operations such as dimension reduction, SConv, SAP, and SLW iteratively. Finally, in the prediction phase, all superpixels pass through the classifier to acquire the final classification map. The SACNN framework helps in capturing the spectral, spatial, and geometric information of HSIs to achieve accurate object recognition.

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Algorithm 1 SACNN Algorithm for HSI Classification

Require: Raw HSI data $\mathbf{H} \in \mathbb{R}^{X \times Y \times B}$ and class labels.
Ensure: Predicted classification results for HSI.

1: Randomly select 5-15 samples of each class from the labeled samples to form a training set, and the rest of the samples form a testing set.
2: Calculate the number of HSI superpixels N_{SP} by Equation (4), and perform the SLIC segmentation algorithm on $\mathbf{H} \in \mathbb{R}^{X \times Y \times B}$.
3: Obtain the SHR $\boldsymbol{V}_i$ of each superpixel by Equation (5) and (6), and obtain the feature cube $F_{\mathcal{T}}^{(n)}$ of labeled samples by Equation (7)-(10).
4: Perform variable size regularization on batch to obtain the regularized feature map $\widetilde{F}_{\mathcal{T}_t}^{(n)}$ by Equation (11) and (12) for batch training.
5: Perform the iterative training of SACNN using the proposed SConv, SAP, SLW, and loss function by Equation(13)-(22).
6: Employ the Adam optimizer to iteratively train for 300 epochs, complete the SACNN parameter learning, and reserve a relatively optimal network.
7: All pixels are classified with the optimal SACNN model, and the predicted classification map of HSI is obtained.

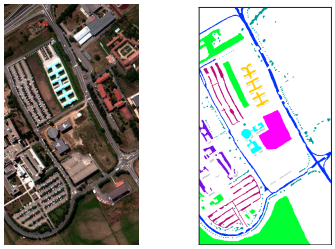


(a) (b)

Fig. 7. YRE dataset: (a) false-color image and (b) ground truth distribution.

TABLE II
LAND COVER CLASSES IN YRE DATASET

Class	Color	Land Cover Type	Number
C1		Building	533
C2		River	5376
C3		Salt marsh	4995
C4		Shallow sea	17550
C5		Deep sea	18677
C6		Intertidal saltwater marsh	2343
C7		Tidal flat	1792
C8		Pond	1787
C9		Sorghum	646
C10		Corn	1509
C11		Lotus root	2719
C12		Aquaculture	8019
C13		Rice	5508
C14		Tamarix chinensis	1220
C15		Freshwater herbaceous marsh	1417
C16		Suaeda salsa	874
C17		Spartina alterniflora	580
C18		Reed	1970
C19		Floodplain	347
C20		Locust	75
		Total	77937



(a) (b)

Fig. 8. PU dataset: (a) false-color image and (b) ground truth distribution.

IV. EXPERIMENTAL SETUP

In this section, three hyperspectral datasets of Yellow River Estuary (YRE), Pavia University (PU), and Indian Pines (IP) are adopted to verify the effectiveness of the proposed SACNN in this article. Class accuracy (CA), overall accuracy (OA), average accuracy (AA), and kappa coefficient (κ) are employed as the quantitative evaluation indicators of classification performance, along with qualitative analysis based on visualization results in the comparative experiments. For each of the three datasets, several random samples from each category are selected to form the training set, while the remaining samples are used for the testing set. To minimize the influence of random factors, each experiment is repeated multiple times to obtain the average values of performance indicators.

A. Datasets

1) Yellow River Estuary: The YRE is a hyperspectral dataset acquired by the visible shortwave infrared hyperspectral camera (AHSI), which is designed by the Shanghai Institute of Technical Physics, Chinese Academy of Sciences, and installed on China's Gaofen-5 satellite platform [47]. The dataset was collected on November 1, 2018, over the YRE in the northeast of Shandong Province, China, and primarily consists of wetland vegetation. The AHSI sensor collected 330 spectral bands, including 150 visible to near-infrared bands (400–1000 nm) and 180 shortwave infrared bands (1000–2500 nm). After removing the bands with poor quality, the remaining 280 bands are used in this study. The image size of YRE is 1400 × 1400, with a spatial resolution of 30 m. The dataset contains 20 ground object categories, and 77 937 pixels are labeled as samples. Fig. 7(a) shows the false-color image of YRE HSI, Fig. 7(b) shows the distribution of labeled samples, and Table II provides information on the number of labeled samples for each category.

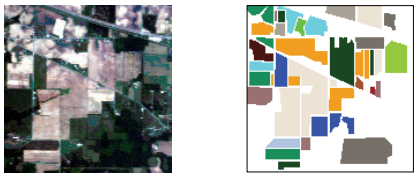
2) Pavia University: The PU dataset was acquired by the reflective optics spectrographic imaging system (ROSIS) at the University of Pavia in northern Italy in 2003. The original dataset consists of 115 spectral bands, and after removing 12 noise bands, 103 bands are retained for experiments. The image size of PU is 610 × 340, with a spatial resolution of 1.3 m. There are a total of 42 776 labeled pixel samples distributed among nine categories. Fig. 8(a) shows the false-color map of PU HSI, Fig. 8(b) shows the distribution of labeled samples, and Table III provides information on the number of labeled samples for each category.

3) Indian Pines: The IP dataset was acquired by the airborne visible infrared imaging spectrometer (AVIRIS) on

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TABLE III
LAND COVER CLASSES IN PU DATASET

Class	Color	Land Cover Type	Number
C1		Asphalt	6631
C2		Meadows	18 649
C3		Gravel	2099
C4		Trees	3064
C5		Painted metal sheets	1345
C6		Bare soil	5029
C7		Bitumen	1330
C8		Self-blocking bricks	3682
C9		Shadows	947
		Total	42776



(a) (b)

Fig. 9. IP dataset: (a) false-color image and (b) ground truth distribution.

TABLE IV
LAND COVER CLASSES IN IP DATASET

Class	Color	Land Cover Type	Number
C1		Stone-steel-towers	95
C2		Hay-windrowed	489
C3		Corn-mintill	834
C4		Soybeans-notill	968
C5		Alfalfa	54
C6		Soybeans-clean	614
C7		Grass-pasture	497
C8		Woods	1294
C9		Buildings-grass-tree-drives	380
C10		Grass-pasture-mowed	26
C11		Corn	234
C12		Oats	20
C13		Corn-notil	1434
C14		Soybeans-mintill	2468
C15		Grass-trees	747
C16		Wheat	212
		Total	10366

June 12, 1992, over northwestern Indiana. The original dataset consists of 224 spectral bands, with 4 zero bands and 20 bands that cannot be reflected by water being discarded, leaving 200 bands for experiments. The image size of IP is 145 × 145, with a spatial resolution of 20 m. There are a total of 10 366 labeled pixel samples distributed among 16 categories. Fig. 9(a) shows the false-color map of IP HSI, Fig. 9(b) shows the distribution of labeled samples, and Table IV provides information on the number of labeled samples for each category.

B. Experimental Setup

In order to effectively verify the classification performance of the proposed SACNN model in this article, one method of traditional machine learning and eight methods of deep learning are employed in the experiment for comparison. The detailed description of comparison methods is as follows.

1) Two-dimensional Gabor-based SVM (2DG-SVM) method extracts the features through 2-D Gabor filters, reduces the dimension using principal component analysis (PCA), and inputs them into the SVM for classification.
2) Two-dimensional CNN (2DCNN) method inputs the original HSI data into the CNN with 2-D standard convolution kernels for classification. The network consists of three convolutional modules and a fully connected layer [48].
3) The multibranch CNN (MB-CNN) method is a fusion network formed by merging multiple branches of common CNN. The network introduces a 1 × 1 convolutional layer and L2 regularization to extract the features of HSI data [49].
4) Spectral Former (SpectralFormer) method is an HSI classification network based on the transformer model. The network extracts features such as texture and edge details for classification by enhancing the adjacent spectral information and transferring the "memory" across layers [50].
5) Bottleneck spatial-spectral transformer (BS2T) method is also based on the transformer model, which employs the bottleneck transformer module to extract spatial and spectral features and introduces spectral information into the original multihead attention [51].
6) Attention adaptive spatial-spectral kernel-ResNet (A2S2K-ResNet) method captures the discriminative features in an end-to-end manner through spectral attention and uses 3-D residual block and 3-D convolution kernel to jointly extract spectral-spatial features [52].
7) Ensemble of multiple CNN (MECNN) method integrates multiple superpixel smoothing CNN classifiers, utilizes stacked AEs to reduce the data dimension, and investigates the diversity among classifiers to find the best combination [41].
8) Multitask deep learning for the open world (MDL4OW) is a classification method under the few-shot context. It simultaneously conducts classification and reconstruction in the open world where unknown classes may exist, based on a multitask deep-learning framework [53].
9) CNN-enhanced GCN (CEGCN) introduces a heterogeneous network with dual branches of CNN and GCN. GCN branch is employed to extract superpixel-level spatial features, and the local information of the CNN branch is used to simultaneously generate complementary spectral-spatial features at pixel and superpixel levels [54].

For the experimental settings of all comparison methods, deep-learning methods set the input window size to 5, and the other comparison methods follow their original papers. For the training settings of SACNN, Adam optimizer [55] is adopted to update the parameters for 300 training epochs, and the initial learning rate is set to 0.001. Since the input of SACNN is different from other deep-learning methods, $L = W = 5$ is set in (4) for a fair comparison.

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TABLE V

ABLATION EXPERIMENTAL RESULTS OF SCONV, SAP, AND SLW ON THREE DATASETS

SConv	SAP	Ω	YRE		PU		IP	
			OA (%)	κ	OA (%)	κ	OA (%)	κ
✗	✗	✗	85.79	0.83	76.20	0.75	63.49	0.59
✗	✗	✓	86.28	0.84	77.31	0.76	66.04	0.62
✗	✓	✗	86.42	0.84	78.84	0.76	70.69	0.68
✗	✓	✓	87.16	0.85	80.32	0.77	73.05	0.70
✓	✗	✗	88.39	0.86	82.97	0.81	77.65	0.74
✓	✗	✓	90.65	0.89	86.26	0.83	80.29	0.77
✓	✓	✗	91.10	0.90	88.43	0.86	83.65	0.81
✓	✓	✓	**93.11**	**0.92**	**90.96**	**0.88**	**85.04**	**0.83**

V. EXPERIMENTAL RESULTS

A. Ablation Experiments

According to the spatial distribution characteristics of superpixel regularization proposed in this study, the SACNN method improves the original convolution operation and pooling operation. Based on this, we propose the new operators of SConv and SAP. In addition, to address the issue of the same input corresponding to different labels, the SLW is designed. To verify the effectiveness of the aforementioned improvements in the SACNN model, we conduct the ablation experiments on three datasets using different network module settings, as shown in Table V. We randomly select 15 labeled samples per class to form the training set and obtain the average value of performance indicators after ten repetitions. In the table, "✗" denotes utilizing the traditional convolution, GAP or cross-entropy (CE) loss, and "✓" denotes employing the proposed SConv, SAP or SLW (Ω). Comparing the first and fifth rows as well as the fourth and last rows in Table V, it is obvious that SConv has better spectral, spatial, and geometric feature extraction capabilities compared to the traditional convolution. Based on superpixel segmentation and regularization, it learns more comprehensive and significant information about different ground objects. From the first and third rows as well as the sixth and last rows in Table V, it can be seen that SAP has superior feature aggregation ability for SHRs. In addition, from the first and second rows as well as the seventh and last rows, it is noticed that SLW is able to adjust the classification weights at different positions, tradeoff the respective classification probabilities, and obtain superior recognition results. Therefore, the results of ablation experiments consistently indicate that our proposed SConv, SAP, and SLW modules can effectively improve the feature learning and classification performance of the SACNN model. From the experimental results, the "SConv + SAP + Ω" (the last row of Table V) network (SACNN) achieves the best classification accuracy and consistency on three datasets.

B. Parameter Analysis

In order to explore the influence of parameters L and W in (4) on the classification accuracy of the SACNN method, the parameter analysis is carried out on three datasets. For each dataset, 15 labeled samples per class are randomly selected to form the training set and the average value of performance indicators after ten repetitions is obtained for evaluation. Different combinations of parameters L and W are set among 3, 5, and 7 to explore the performance relationships, which are commonly adopted in superpixel segmentation and CNNs, as shown in Table VI. As seen in the table, when both L and W are set to 5, the SACNN method achieves the highest classification accuracy on three datasets. This window size is more suitable for segmenting the superpixels and extracting the spatial-spectral features, which is conducive to learning the deep structure-adaptive convolutional features. On one hand, if L and W are set too small (e.g., $L = W = 3$), the number of superpixels N_{sp} obtained by segmentation would be too large, resulting in less information of homogeneous features in each superpixel, which can negatively impact the recognition of ground objects. On the other hand, if L and W are set too large (e.g., $L = W = 7$), the homogeneity within each superpixel would be reduced and irrelevant noise would be introduced, which can also affect the classification accuracy. In the parameter experiments, different values of L and W are considered simultaneously. When $L = 3$ and $W = 5$ [equivalent to the number of superpixel segmentation when $L = 5$ and $W = 3$ according to (4)], the classification accuracy of SACNN method falls between $L = W = 5$ and $L = W = 3$. When $L = 5$ and $W = 7$ (equivalent to the number of superpixel segmentation when $L = 7$ and $W = 5$), the SACNN method's accuracy falls between $L = W = 5$ and $L = W = 7$. Therefore, we set $L = W = 5$ for conducting related classification experiments in order to obtain a more stable classification performance.

TABLE VI

EXPERIMENTAL RESULTS FOR L AND W PARAMETER SETTINGS

L, W	YRE		PU		IP	
	OA (%)	κ	OA (%)	κ	OA (%)	κ
L=3, W=3	91.50	0.88	87.85	0.85	82.75	0.80
L=3, W=5	92.56	0.90	89.21	0.86	83.52	0.81
L=5, W=5	**93.11**	**0.92**	**90.96**	**0.88**	**85.04**	**0.83**
L=5, W=7	91.06	0.89	89.13	0.87	83.25	0.81
L=7, W=7	90.34	0.88	88.05	0.86	82.30	0.79

TABLE VII

EXPERIMENTAL RESULTS FOR COMPACTNESS PARAMETER SETTINGS

Compactness	YRE		PU		IP	
	OA (%)	κ	OA (%)	κ	OA (%)	κ
0.01	92.24	0.91	90.07	0.87	84.57	0.82
0.1	92.53	**0.92**	90.46	**0.88**	84.72	0.82
1	92.94	**0.92**	90.73	**0.88**	84.89	**0.83**
10	**93.11**	**0.92**	**90.96**	**0.88**	**85.04**	**0.83**
100	92.97	**0.92**	90.23	0.87	84.87	0.82

In addition, experiments are conducted to explore the impact of different hyperparameters of SLIC superpixel segmentation on the SACNN classification performance. Specifically, the compactness parameter is varied to examine its effect on the regularity and boundary consistency of superpixels. Higher values of compactness result in more regular and square-shaped superpixels, while lower values produce more irregular superpixels with improved boundary consistency. To ensure reliable experimental results, we repeated the experiment ten times with different compactness values (0.01, 0.1, 1, 10, and 100) on three datasets. The average of these ten results was taken as the final experimental result. Table VII demonstrates that the OA and kappa coefficient obtained from the three

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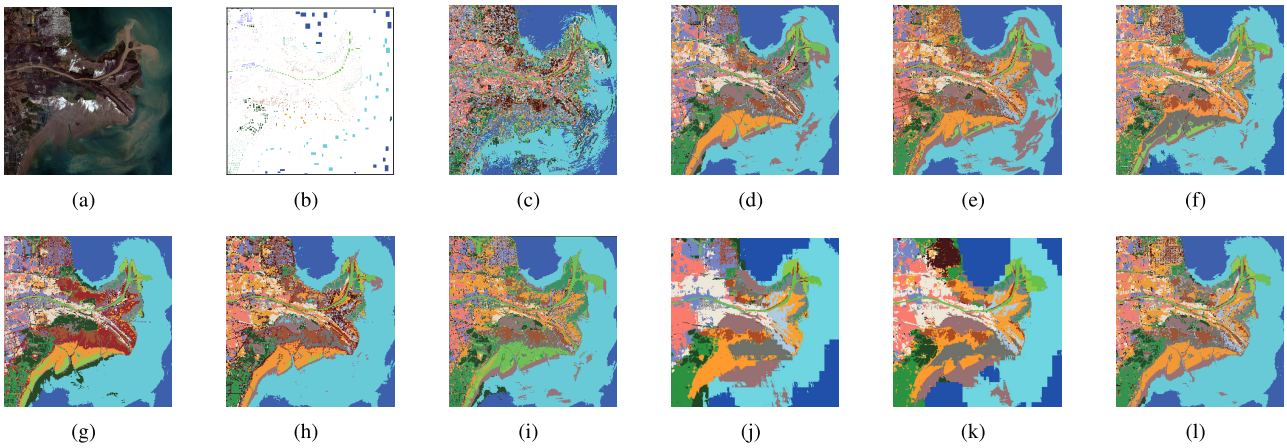


Fig. 10. Classification maps of different methods on the YRE dataset: (a) false-color image, (b) ground truth, (c) 2DG-SVM, (d) 2DCNN, (e) MB-CNN, (f) SpectralFormer, (g) BS2T, (h) A2S2K-ResNet, (i) MECNN, (j) MDL4OW, (k) CEGCN, and (l) SACNN.

TABLE VIII
CLASSIFICATION ACCURACY (%) AND KAPPA COEFFICIENT FOR YRE DATASET WITH 15 LABELED SAMPLES PER CLASS

Class	2DG-SVM	2DCNN	MB-CNN	SpectralFormer	BS2T	A2S2K-ResNet	MECNN	MDL4OW	CEGCN	SACNN
C1	62.39	60.06	75.91	62.31	75.81	51.25	75.72	75.23	**76.99**	**80.30**
C2	89.58	94.67	94.37	98.90	**99.85**	96.90	98.75	95.65	99.18	**99.78**
C3	75.83	58.87	59.07	83.10	70.92	89.20	83.28	89.68	**89.79**	**93.39**
C4	79.42	84.31	86.00	86.63	86.03	**93.09**	92.94	90.00	90.73	**96.30**
C5	76.64	93.74	94.89	**99.12**	98.13	96.33	97.69	**98.75**	95.38	96.99
C6	75.17	80.38	65.37	**83.95**	81.07	67.44	68.52	**84.51**	77.83	79.47
C7	49.80	58.84	**80.92**	64.77	74.07	66.93	**84.35**	55.69	69.21	61.22
C8	67.66	82.32	84.97	80.23	86.70	86.97	86.30	86.63	**93.65**	**95.86**
C9	85.66	76.63	77.24	64.40	85.96	87.50	79.72	87.34	**89.92**	**91.18**
C10	43.72	71.96	60.46	67.84	**92.91**	82.58	86.22	**98.87**	85.91	80.98
C11	67.37	72.96	84.13	**84.93**	75.55	81.41	81.73	75.91	84.59	**85.07**
C12	64.83	76.43	84.50	78.36	74.11	78.00	82.33	79.55	**93.37**	**93.39**
C13	71.38	68.83	82.43	85.58	79.91	78.12	81.08	84.90	**90.70**	**93.83**
C14	67.45	84.22	89.52	81.54	91.88	90.29	84.41	91.07	**92.28**	**92.95**
C15	70.81	83.41	86.95	**94.97**	89.54	**93.37**	89.09	89.40	88.01	92.87
C16	82.37	82.29	90.88	84.92	87.10	90.05	89.07	**92.83**	91.74	**95.65**
C17	85.28	78.68	80.37	89.78	**99.47**	98.41	89.35	96.21	97.91	**98.97**
C18	45.80	50.70	70.18	68.50	71.65	75.46	**77.66**	72.08	**85.26**	71.88
C19	80.97	55.22	57.82	73.82	79.40	77.18	**86.49**	79.83	84.76	**87.61**
C20	77.03	94.74	75.00	76.50	93.65	85.26	**95.08**	**98.67**	86.00	80.00
OA(%)	73.79	81.08	84.66	87.41	87.72	88.12	88.57	89.34	92.14	**93.11**
AA(%)	70.96	75.46	79.05	80.52	84.53	83.28	85.48	86.14	88.16	**88.38**
κ	0.77	0.79	0.84	0.85	0.86	0.86	0.87	0.88	0.91	**0.92**

datasets remain relatively close across different compactness values. This indicates the robustness of our proposed method, attributed to the innovative acquisition of SHR and variable size regularization. It also highlights the strong adaptability of SConv, SAP, and SLW in our approach. In the subsequent experiments, the compactness of SLIC segmentation is set to 10 for stable classification performance.

C. Comparison Experiments

In accordance with the experimental settings of comparison methods, the training and testing samples are randomly selected from each dataset. Specifically, 15 randomly labeled samples per class are selected to form the training set, and this process is repeated 10 times to mitigate the impact of random factors. The average values of CA, OA, AA, and κ are calculated for the performance comparison. In addition, one traditional machine-learning method and eight deep-learning methods are adopted as the comparison methods on three datasets, and the experimental results are analyzed both quantitatively and qualitatively.

1) Yellow River Estuary: Table VIII presents the results of classification obtained by SACNN and other comparison methods on the YRE dataset. The best and second-best CA as well as the best OA are highlighted in bold in each row. As demonstrated in the table, SACNN achieves better classification results in most classes as well as OA. 2DG-SVM exhibits the lowest classification accuracy, which employs hand-crafted feature extraction for classification. MB-CNN performs better than 2DCNN with its multibranch fusion design. However, transformer-based methods such as SpectralFormer and BS2T have limitations in classification accuracy, considering their complex network structure and small number of training samples. Although BS2T incorporates the BS2T and spectral attention, its improvement in accuracy compared to SpectralFormer is not significant. A2S2K-ResNet, which utilizes adaptive learning with 3-D convolution kernels and 3-D residual blocks, shows improved accuracy compared to 2DCNN and MB-CNN. MECNN integrates four CNN classifiers with superpixel smoothing and searches for the optimal features, achieving a similar OA to A2S2K-ResNet.

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TABLE IX
CLASSIFICATION ACCURACY (%) AND KAPPA COEFFICIENT FOR PU DATASET WITH 15 LABELED SAMPLES PER CLASS

Class	2DG-SVM	2DCNN	MB-CNN	SpectralFormer	BS2T	A2S2K-ResNet	MECNN	MD4LOW	CEGCN	SACNN
C1	54.82	77.09	68.52	64.27	**94.85**	74.49	73.76	84.04	87.13	**89.71**
C2	60.76	83.13	89.82	84.70	**99.77**	**94.70**	78.88	81.17	92.37	90.77
C3	61.16	86.00	62.10	63.38	**86.66**	60.68	60.98	77.22	76.75	**86.95**
C4	64.27	87.55	70.87	95.72	77.38	71.37	88.32	93.69	**95.87**	**96.56**
C5	76.90	99.63	72.66	**100.00**	86.07	79.44	89.40	98.88	98.54	**100.00**
C6	57.78	19.67	61.76	64.67	72.65	**86.40**	79.94	78.55	82.51	**85.33**
C7	65.22	94.06	70.04	78.69	**98.05**	68.44	91.42	94.59	90.93	**96.54**
C8	55.06	59.29	63.80	64.26	73.96	47.10	47.72	78.24	**81.95**	**83.89**
C9	74.52	99.47	71.57	**99.74**	86.21	62.30	**99.89**	96.62	99.53	94.08
OA(%)	60.22	74.36	76.70	77.76	87.34	78.67	78.98	83.21	87.99	**90.96**
AA(%)	63.39	78.43	70.13	79.47	86.18	71.66	80.04	87.00	89.51	**91.54**
κ	0.60	0.67	0.71	0.74	0.84	0.72	0.73	0.78	0.85	**0.88**

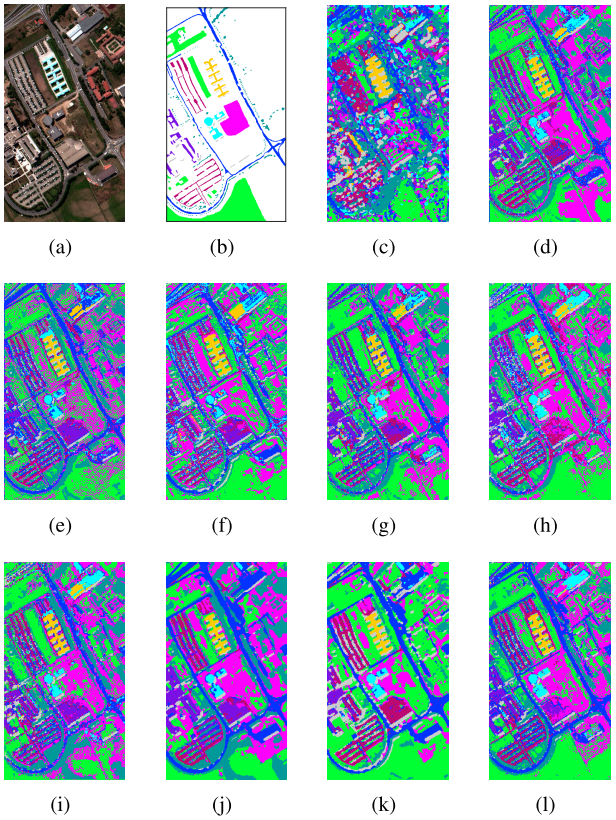


Fig. 11. Classification maps of different methods on the PU dataset: (a) false-color image, (b) ground truth, (c) 2DG-SVM, (d) 2DCNN, (e) MB-CNN, (f) SpectralFormer, (g) BS2T, (h) A2S2K-ResNet, (i) MECNN, (j) MDL4OW, (k) CEGCN, and (l) SACNN.

For the few-shot classification methods, MDL4OW employs a multitasking framework to perform classification and reconstruction simultaneously, also obtaining similar accuracy. Since CEGCN combines CNN and GCN to extract local and global features, its classification performance is better than MDL4OW. Finally, SACNN realizes the structure-adaptive deep spatial-spectral feature learning and achieves the best overall classification accuracy of 93.11% and kappa coefficient of 0.92 with only 15 training samples per class in the YRE dataset, surpassing other comparison methods. In addition, Fig. 10 shows the classification maps of each method in a single experiment. It shows that SACNN produces the classification map with better class boundary consistency, better object shape compactness, and lower object recognition confusion. In conclusion, the experimental results validate the effectiveness and reliability of the proposed SACNN method for HSI classification.

2) Pavia University: Table IX presents the classification results of SACNN and other comparison methods on the PU dataset. The best and second-best CA as well as the best OA are highlighted in bold in each row. According to the table, SACNN outperforms other methods for most classes, particularly in C4 (trees), C5 (painted metal sheets), and C7 (bitumen), achieving a CA of over 95%. Overall, SACNN achieves the highest OA of 90.96%, AA of 91.54%, and kappa coefficient of 0.88. The overall classification accuracy of the A2S2K-ResNet method is slightly lower than that of MECNN. These two methods incorporate the attention feature extraction and superpixel feature fusion modules, respectively, obtaining better results compared to 2DCNN and MB-CNN. Between the two transformer-based methods, BS2T shows better classification accuracy and consistency than SpectralFormer. Although MD4LOW and CEGCN do not achieve more optimal or suboptimal results in CA, they have similar overall performance to the BS2T, with higher OA and kappa coefficients than most comparison methods. In addition, Fig. 11 shows the classification maps of each method in a single experiment. Qualitative analysis reveals that SACNN learns better spectral, spatial, and geometric features, and its classification map has smoother class boundaries and more accurate spatial distribution compared to other methods. The classification boundaries of MB-CNN methods are relatively more fragmented, and the classification boundaries of the 2DG-SVM method are not accurate enough and show obvious confusion between ground objects.

3) Indian Pines: Table X presents the classification maps of SACNN and other comparison methods on the IP dataset. The best and second-best CA as well as the best OA are highlighted in bold in each row. SACNN achieves the highest OA of 85.04%, AA of 89.51%, and kappa coefficient of 0.83. It also shows better CA for most classes, particularly in C9 (buildings-grass-tree-drives) with a CA of 100%. 2DCNN, MB-CNN, SpectralFormer, and MECNN methods show similar overall classification accuracy, but SpectralFormer exhibits obviously different accuracy among various classes, indicating unbalanced recognition performance. For instance,

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TABLE X
CLASSIFICATION ACCURACY (%) AND KAPPA COEFFICIENT FOR IP DATASET WITH 15 LABELED SAMPLES PER CLASS

Class	2DG-SVM	2DCNN	MB-CNN	SpectralFormer	BS2T	A2S2K-ResNet	MECNN	MDL4OW	CEGCN	SACNN
C1	**94.00**	84.40	90.20	93.54	76.85	50.12	89.47	93.75	92.90	**95.65**
C2	**91.22**	83.13	88.16	43.41	78.95	74.39	87.64	77.84	62.59	**90.45**
C3	61.24	58.50	70.06	54.41	63.77	**87.42**	64.66	65.30	69.83	**77.95**
C4	53.41	60.83	69.13	61.71	81.48	75.60	63.98	92.19	**95.74**	**96.20**
C5	**94.00**	84.03	90.20	71.68	88.68	91.93	79.07	87.99	**92.50**	84.89
C6	67.72	64.30	62.84	78.99	**89.16**	80.28	69.56	77.64	76.4	**89.32**
C7	68.23	76.83	80.34	**100.00**	59.60	69.51	80.08	94.82	91.25	**96.43**
C8	76.86	79.49	75.98	97.23	86.48	**99.56**	91.42	96.05	89.53	**98.12**
C9	73.82	73.03	83.09	**100.00**	52.15	61.11	87.89	**100.00**	98.41	**100.00**
C10	**94.00**	83.63	90.20	66.84	75.30	86.76	**95.00**	77.67	92.24	87.65
C11	88.57	77.99	83.36	57.61	63.78	**94.93**	72.48	78.41	**96.72**	86.40
C12	94.00	84.40	90.20	41.38	77.30	82.11	**97.00**	86.81	**99.47**	65.09
C13	35.39	51.18	57.08	**99.52**	49.03	66.67	50.35	73.57	57.36	**79.51**
C14	34.07	48.81	44.74	86.00	52.62	**97.62**	55.80	89.96	90.95	**93.60**
C15	52.74	81.54	86.72	46.81	78.45	40.22	79.11	89.20	**94.87**	**98.45**
C16	93.51	84.02	89.73	**99.61**	60.59	62.17	90.09	**96.21**	95.75	92.47
OA(%)	56.29	64.04	66.57	63.85	75.42	77.27	68.08	80.07	82.95	**85.04**
AA(%)	73.30	73.51	78.25	74.92	70.89	76.27	78.35	86.09	87.29	**89.51**
κ	0.57	0.63	0.64	0.59	0.72	0.74	0.62	0.77	0.80	**0.83**

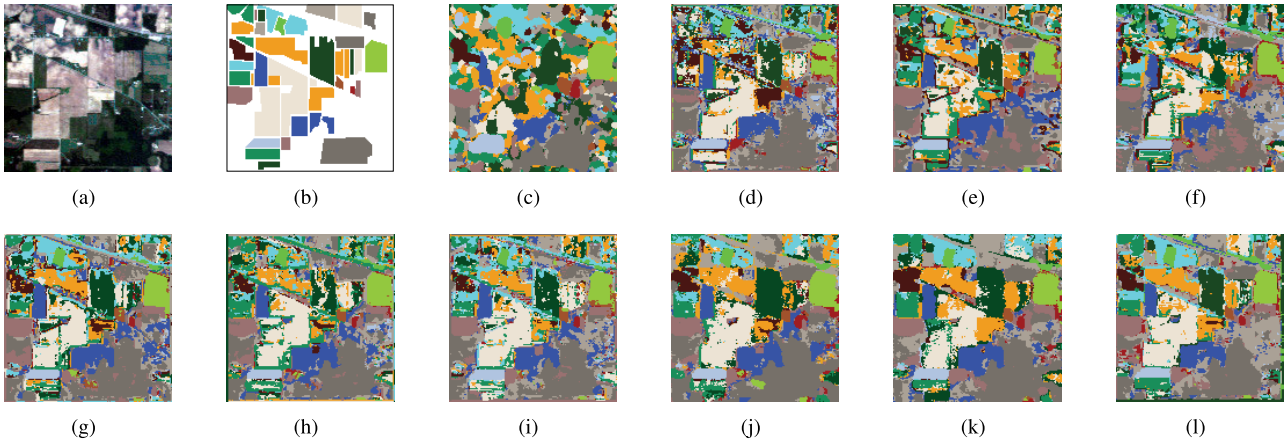
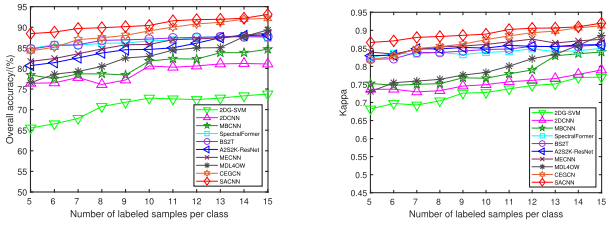


Fig. 12. Classification maps of different methods on the IP dataset: (a) false-color image, (b) ground truth, (c) 2DG-SVM, (d) 2DCNN, (e) MB-CNN, (f) SpectralFormer, (g) BS2T, (h) A2S2K-ResNet, (i) MECNN, (j) MDL4OW, (k) CEGCN, and (l) SACNN.

higher accuracy is achieved in C7 (Grass-pasture) and C9, while lower accuracy is obtained in C2 (Hay-windrowed), C12 (Oats), and C15 (Grass-trees). The classification results of A2S2K-ResNet and BS2T methods are relatively close to each other. For MDL4OW and CEGCN, their OA and AA exceed 80% and 85%, respectively. Fig. 12 shows the classification maps of each method in a single experiment. It reveals that the classification map of SACNN shows a distribution of predicted objects that is closer to the ground truth, with better discrimination and interpretation. SpectralFormer method tends to mix misjudgments of other ground objects in the continuous recognition of large-region ground objects. The MDL4OW and CEGCN methods obtain relatively clear classification maps but do not perform well at the boundaries of fuzzy objects.

D. Performance Under Different Sample Sizes

In order to further verify the classification performance of the proposed method in case of a small sample size, the accuracy of SACNN and comparison methods under different sample sizes is analyzed. The experiments are conducted in the same environment, using 5–15 labeled samples per class for training and repeating ten times to obtain the average OA and kappa coefficient.

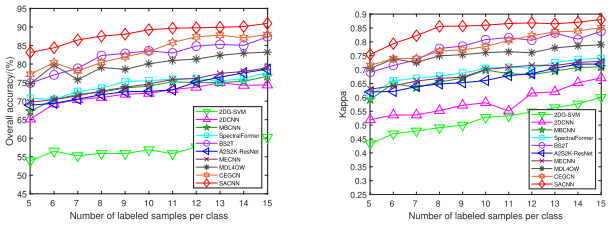


(a) (b)

Fig. 13. Classification results of different methods on the YRE dataset with increasing sample size: (a) OA and (b) kappa.

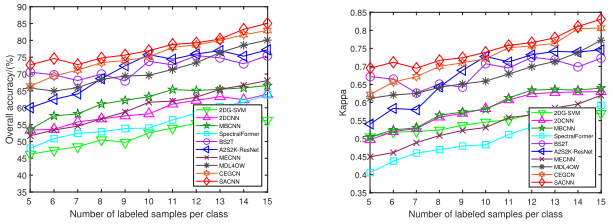
Figs. 13–15 show the changes in classification results of different methods on three datasets with varying numbers of training samples. From the figures, it can be observed that as the number of training samples increases, the overall classification accuracy improves due to more comprehensive feature extraction and relationship fitting. The proposed SACNN method achieves the best classification results across different sample sizes in three datasets, particularly in cases with limited samples, showing its ability to effectively extract the spectral, spatial, and geometric features and

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(a) (b)

Fig. 14. Classification results of different methods on the PU dataset with increasing sample size: (a) OA and (b) kappa.



(a) (b)

Fig. 15. Classification results of different methods on the IP dataset with increasing sample size: (a) OA and (b) kappa.

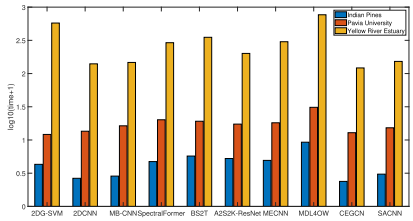


Fig. 16. Total inferring time (seconds) of comparison methods on the three datasets.

adaptively learn from the superpixel structures through deep CNN. The recognition performance of 2DG-SVM methods is generally poor, due to the insufficient feature extraction. The classification accuracy of 2DCNN, MB-CNN, and MECNN methods falls in the moderate range for small-sample classification, with MECNN showing slightly superior performance due to the integration of multiple CNN classifiers. SpectralFormer method exhibits relatively better classification results on the YRE and PU datasets compared to the IP dataset, which is possibly related to the differences in ground object distribution. The BS2T, A2S2K-ResNet, and MDL4OW methods are more sensitive to the number of training samples, which is particularly evident in the IP dataset, indicating that the classification performance of deep-learning methods depends on the size of training samples. As for CEGCN, it maintains classification performance that is second only to SACNN across all three datasets. In general, the SACNN method achieves the best classification results, getting benefits from the optimized SConv and pooling operators and the designed SLW for robust feature extraction.

E. Model Inferring Time Analysis

In practical applications, the time it takes for a classification model to make predictions (referred to as "inferring time") is often more important than the time it takes to train the model. Therefore, we conduct experiments to compare the inferring time of different classification methods on three datasets using a Tesla P100 GPU with 16GB of memory. As shown in Fig. 16, the logarithmic operations on the inferring time are performed to facilitate the comparative analysis of full-image prediction. From the figure, it can be observed that the CEGCN has the shortest inferring time, due to its ability to utilize structural information and reduce computational iterations. The 2DCNN, MB-CNN, and SACNN methods have relatively similar inferring times. The SACNN accurately extracts the comprehensive spatial-spectral features of irregular superpixels based on CNN, without significantly increasing the number of parameters and computational complexity. The inferring time of A2S2K-ResNet and MECNN methods has increased compared to 2DCNN and MB-CNN due to the addition of attention mechanism and feature fusion strategy. In addition, the inferring time of BS2T and SpectralFormer methods is slightly higher than that of CNN-based methods, possibly related to the calculation of feature embedding and multihead self-attention. The 2DG-SVM method requires more time to calculate the Gabor features, and the MDL4OW has additional computation cost and the most inference time, due to the reconstruction task involved in the model. Our SACNN method, which designs innovative convolutional and pooling operators as well as a reasonable architecture, achieves the best classification accuracy with short inferring time, making it an optimal choice for HSI classification performance.

VI. Conclusion

In this article, we propose a SACNN for HSI classification. The goal of SACNN is to address the issue of limited labeled samples, extensive training requirements, and complicated spatial features, which can impact classification performance. Hence, we combine the prior knowledge of superpixel segmentation with a deep CNN model for robust feature learning. SACNN designs the key components of structure-adaptive convolutional and pooling operators as well as SLW, which enhance the ability of spatial feature extraction and reduce the number of parameters. Experimental results demonstrate that SACNN effectively extracts deep features from spectral, spatial, and geometric structures and achieves better accuracy and consistency than other state-of-the-art methods for HSI classification. Furthermore, SACNN better represents ground objects by mining the data characteristics and distribution patterns of HSI, achieving significant performance improvement for object recognition. In the future, we will investigate how to further utilize multiscale superpixel segmentation and self-supervised contrastive learning to extract spatial-spectral fusion features in HSI based on the method proposed in this article.

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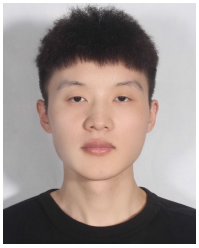
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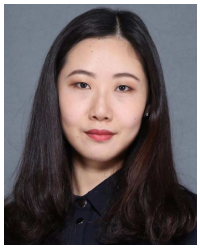
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